

STPI Master Dossier: Scientist 'B'

Interview Preparation

Candidate: Avanish Kumar Gupta

Current Role: Associate Software Engineer, Berkadia (C++, Java, SQL)

Prior Experience: Project Trainee, STMicroelectronics

Interview Date: July 19, 2026 | **Reporting Time:** 9:00 AM | **Venue:** STPI HQ, New Delhi

CBT Score: 96/150 (Top Tier)

SECTION 1: Organizational Genesis & Core Mandate

Software Technology Parks of India (STPI) was established in **1991** as an autonomous society under the **Ministry of Electronics and Information Technology (MeitY)**.

- **The Big Shift:** Historically, STPI was created to boost IT/ITeS exports through statutory single-window clearances. Today, its mandate has aggressively shifted toward making India a **"Software Product Nation"** (aligning with the National Policy on Software Products 2019) by incubating deep-tech startups and fostering Intellectual Property (IP) creation.
- **Scale and Reach:** STPI operates **73 centres** across India. Crucially, **65 of these are in Tier-II and Tier-III cities** (like Bhopal, Bhubaneswar, Dehradun). STPI's primary goal is to democratize tech growth so it isn't limited to Bengaluru, Hyderabad, and Pune.

SECTION 2: The Core Schemes (Must-Know)

As a Scientist 'B', you will be directly or indirectly evaluating, managing, or mentoring initiatives under these schemes. You must know these acronyms:

1. NGIS (Next Generation Incubation Scheme)

- **What it is:** A ₹95.03 Crore futuristic incubation scheme focused on 12 Tier-II locations (Agartala, Bhilai, Bhopal, Bhubaneswar, Dehradun, Guwahati, Jaipur, Lucknow, Prayagraj, Mohali, Patna, Vijayawada).
- **Target:** To support 300 innovative software product startups over 3 years.
- **Funding:** Provides a seed fund of up to **₹25 Lakh** per startup and ₹10,000/month intern stipends during pre-incubation.

2. CHUNAUTI (Challenge Hunt Under NGIS for Advanced Uninhibited Technology Intervention)

- **What it is:** The actual screening/hackathon process used to select startups for the NGIS program.

- **Note:** STPI regularly launches iterations of this (e.g., CHUNAUTI 8.0/9.0) focusing on emerging tech, EdTech, FinTech, and AgriTech.

3. LEAP AHEAD (Launchpad for Entrepreneurs towards Accelerated growth and Pioneering AHEAD)

- **What it is:** Recently launched its 2nd Edition (late 2024/2025). This is for startups in the *scaling or growth stage* (not just ideation).
- **Benefits:** Offers intensive 3-month mentorship, networking with top global investors, and potential funding of up to **₹1 Crore**.

4. GENESIS (Gen-Next Support for Innovative Startups)

- **What it is:** A massive initiative with a ₹490 Crore outlay aimed at nurturing over **1,600 deep-tech startups** specifically in Tier-II and Tier-III cities over the next five years.

5. STP & EHTP Schemes

- **STP (Software Technology Park):** Provides 100% customs duty exemption on capital goods and allows 100% FDI for software exporters.
- **EHTP (Electronics Hardware Technology Park):** The hardware equivalent, designed to boost the ESDM (Electronic System Design and Manufacturing) sector.

SECTION 3: The 24+ Centres of Entrepreneurship (CoEs)

STPI is building domain-specific hubs across India. You don't need to memorize all of them, but memorize the ones that align with your background (highlighted):

1. **Electropreneur Park (New Delhi & Bhubaneswar):** Focused on Electronics and Semiconductors (ESDM). *(Highly relevant to your STMicroelectronics background).*
2. **IoT OpenLab (Bengaluru):** Internet of Things.
3. **FinBlue (Chennai):** FinTech and Blockchain.
4. **IMAGE (Hyderabad):** AI, Gaming, VFX, and Computer Vision.
5. **MOTION (Pune):** Autonomous Connected Electric Shared (ACES) Mobility.
6. **APIARY (Gurugram):** Blockchain technology.
7. **FASAL (Akola):** IoT in Agriculture.
8. **KALPATARU (Visakhapatnam):** Industry 4.0, AI/ML, Industrial Robotics.
9. **MedTech (Lucknow):** Medical Electronics & Healthcare.
10. **OctaNE (North East):** A network of 8 CoEs focusing on Animation, Data Analytics, AR/VR, and E-commerce across NE states.

SECTION 4: 2025-2026 Export Data & Achievements (Drop these stats in the interview)

- **Staggering Export Growth:** STPI-registered units facilitated **over ₹10.64 to ₹10.69 lakh**

crore (approx. US\$ 123 Billion) in IT/ITES exports in FY 2024-25, making up roughly 50% of India's total national software exports.

- **Bengaluru Tech Summit 2025:** STPI co-hosted the massive 'STPI IT Exports Awards 2025', recognizing top performers across Tier-II and Tier III regions.
- **Milestone:** STPI is celebrating over 34 years of catalyzing India's digital growth, having supported the filing of over 1,080+ IPRs/Patents and the creation of 2,200+ tech products by its incubated startups.

SECTION 5: The "Avanish Strategy" - How to Ace the Panel

You have a 96/150 CBT score. The panel knows you are theoretically brilliant. The interview is a "stress test" to see if you have practical, administrative, and communication maturity.

Here is exactly how you connect your profile to their needs:

Pitch 1: The STMicroelectronics Advantage (Hardware meets Software)

- **The Hook:** "During my time as a Project Trainee at STMicroelectronics, I worked in the DTIT Group on the 'Manufacturing Solutions-GOBM' module."
- **The STPI Connection:** Connect this directly to STPI's **EHTP Scheme** and the **Electropreneur Park CoE**. Tell the panel: *"I understand the complexities of the ESDM (Electronics System Design & Manufacturing) sector. Having worked inside a major semiconductor firm, I understand the software tools and manufacturing solutions that hardware startups need. I can effectively evaluate proposals and mentor startups coming into STPI's hardware-focused CoEs."*

Pitch 2: The Berkadia Reality Check (Production-Level Software)

- **The Hook:** "Currently, as an Associate Software Engineer at Berkadia, I work 10-hour days in a high-pressure environment, managing end-to-end SDLC using C++, Java, and SQL for enterprise financial applications."
- **The STPI Connection:** STPI is funding startups through **NGIS** and **LEAP AHEAD**. Tell the panel: *"A lot of startups have great ideas but terrible software architecture. Because I write, debug, and optimize production-level C++ and Java code daily, I know what scalable, secure software looks like. As a Scientist 'B', I can technically audit these startups, ensure their codebases are robust, and guide them in building enterprise-grade products rather than just prototypes."*

Pitch 3: Answering the Toughest Question - "Why Government?"

- *Expect this:* "Avanish, you are earning well at a US-based MNC (Berkadia). Why do you want to join STPI as a Scientist 'B'?"
- **The Perfect Answer:** *"Working at Berkadia is fantastic for my technical growth, but I am*

ultimately building tools for one specific financial enterprise. At STPI, the scale of impact is national. Instead of writing code for one company, a Scientist 'B' helps incubate and fund the next 100 software product companies in India through schemes like NGIS and GENESIS. I want to be on the side of the table that builds India's tech ecosystem, leveraging my industry experience to ensure government funds are allocated to technically sound startups."

SECTION 6: Last-Minute Tech Brush-Up

Since you have a 9.60 CGPA, they will expect your fundamentals to be flawless. Review these on your flight:

1. **OOPs Concepts:** Polymorphism, Inheritance, Encapsulation, Abstraction (in C++ and Java).
2. **SQL:** Joins, Indexing (Clustered vs Non-Clustered), Normalization, Stored Procedures.
3. **Data Structures:** Be ready to orally explain the time complexity (Big O) of sorting algorithms, HashMaps, and Binary Search Trees.
4. **AI & Emerging Tech:** You built 'QCGenie' (AI-powered QC assistant). Be prepared to explain the AI model you used, how you trained it, and how such AI can be incubated in STPI's **IMAGE CoE** in Hyderabad.

Confidence is key. You are not a fresher; you are a working software engineer bringing private-sector discipline to a government role. Walk in with a smile, take your time answering, and own the room. Good luck!